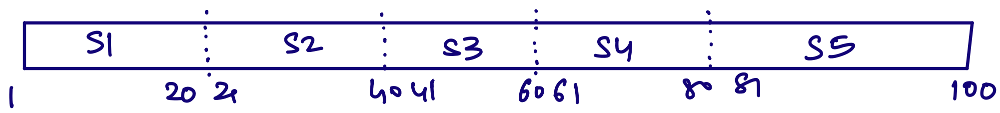
Machine Learning
Trainer: Prashant N.

Cross Validation (CV)

SL → deciding SL
→ standard α values : 0.05 , 0.01, 0.1
→ identify the SL value for given dataset using CV.

dataset containing 100 records



splits *
↓
cv = 5
(hyperparameter)

iteration : 1
s1 → testing set
append (s2, s3, s4, s5) → training set
} → model → score → 0.9
(entire dataset)

iteration: 2
s2 → testing set
append (s1, s3, s4, s5) → training set
} → model → score → 0.95

iteration:
s3 → testing set
append (s1, s2, s4 s5) → training set
} → model → score → 0.95

iteration:
s4 → testing set
append (s1, s2, s3, s5) → training set
} → model → score → 0.95

iteration:
s5 → testing set
append (s1, s2, s3, s4) → training set
} → model → score → 0.95

ADHOC ANALYSIS → Identify SL
→ Discover dataset that achieve SL.

Goal: Discover any hidden pattern or group or association.

Unsupervised Learning

[Given dataset will have only features. No labels

- **Dimensionality Reduction** — compress features
- **Clustering** -or- **Cluster Analysis** — discover hidden groups or categories.
- **Association Rule Mining** -or- **Recommendation Systems.**

Dimensionality Reduction:

The goal of this technique is to COMPRESS feature columns.

Feature Engineering

Feature Elimination -or- **Feature Selection**

- Corr analysis → Regression
- OLS → Regression
- RFE → Regression & Classif^{n}
- SFM → Regression & Classif^{n}

Feature Reduction Technique

- PCA [Principal Component Analysis]

{ Eigen Vector Representation of the feature set/array. }

Set[features] → PCA → compressed features (principal components) → train test split

label → train test split

train test split → train split → ML algo → model

train test split → test split → model → s